Observation Theory

Geometry, Distortion, and Reliability as Properties of Observation

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Abstract

We unify and operationalize a family of results under one thesis—*Observation Theory*: *operational* geometry, distortion, and reliability are not properties of the object alone but of the *relation* among object, observer (a consumer and its output metric), channel, and budget. From a second-order expansion of the downstream metric we recover the leading-order controlling quantity—the reconstruction error read through the consumer, $d_{\mathcal{O}} = \mathbb{E}_x \operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x))$, reducing under stated conditions to $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$—with Σ_δ the error second moment and $P_{\mathcal{C}} = \mathbb{E}[J^\top GJ]$ the output-metric-weighted sensitivity. At root this is a *distinguishability*: $P_{\mathcal{C}} = J^\top GJ$ is the pullback of the consumer's output metric G (the Fisher metric for KL and kindred statistical divergences), so each observer induces a geometry and a quotient $X/\sim_{\mathcal{C}}$ of the representation by what it cannot tell apart. This weighted-quadratic fidelity is classical: it is the indirect/remote source-coding distortion of Dobrushin–Tsybakov and Wolf–Ziv, reduces to Shannon's Gaussian rate–distortion by Berger's change of variables, and is Hessian-weighted loss-aware compression (OBD/OBS through GPTQ) written as a metric. Our contribution is the *middle term*: $P_{\mathcal{C}}$ is a geometric, PSD, water-fillable operator that is *recoverable from black-box consumer queries under a specified observation distribution and output metric*, sitting between Shannon's Gaussian Euclidean-MSE specialization ($P_{\mathcal{C}} = I$) and the information bottleneck. We make that middle term testable and test it, under sealed, before-run registrations with ex-ante bars: distortion—$\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ governs downstream error, demonstrated and replicated across softmax attention, retrieval ranking, and gradient optimization, with a reconstruction-identical code *flipping* its verdict when the consumer changes; identifiability—blind planted-subspace recovery at 16× chance, validated against ground truth; capacity—an extreme-value vacuity threshold that locates retrieval death to a few percent across 14 corpora; and rate—a fixed-budget verdict that *inverts* under budget-matched observation on real embedding manifolds. The analytic consumers are *synthetic* by design (planted read operators, chosen query banks, constructed Hessians), including an $H{=}I$ control where reconstruction becomes exactly optimal; the one real-substrate face is the budget inversion, and hardening on trained-model consumers is the decisive open validation. A fifth conjecture—a density quotient restoring the invariant in non-spectral retrieval—is *refuted* across four prospective tests and carried as a boundary. For the fixed (or averaged) weighted-quadratic model we prove achievability and converse of the consumer-relative rate–distortion function for a general source (Appendix A), and for discrete sources exhibit a consumer-lossless floor below the source entropy. We do *not* claim Shannon was incomplete—his framework already admits any fidelity criterion, and $R_{\mathcal{C}}(D)$ is a classical rate–distortion problem once the consumer's distortion is chosen. Euclidean reconstruction is its *isotropic-observer* corner ($P_{\mathcal{C}} = I$); Observation Theory's contribution is to make that distortion measure an *identifiable geometric object*, recover it from the consumer, and show adversarially that changing it changes which representation is optimal.

1 Introduction

Shannon fixed a distortion measure and asked for the fewest bits meeting it [1, 2]; Gauss fixed the same measure and asked for the least-squares estimate [3]. This paper reads their common assumption as the variable of interest: the error that matters is not the one a code makes but the one a consumer *reads*, and different consumers read different subspaces of the same error. Once the distortion is the task's, Euclidean reconstruction is one corner—the isotropic observer $P_{\mathcal{C}} = I$—and any anisotropic or rank-deficient $P_{\mathcal{C}}$ reads the error differently.

The idea that one should code a source to serve a *functional* of it, not the source, is not new: it is the indirect (remote) rate–distortion problem [5, 6, 7]; its lossless, discrete form is coding for computing and graph entropy [8, 9]; its modern face is the rate–distortion–perception tradeoff [10] and task-oriented / semantic communication [11]; and its deployed form, for neural networks, is loss-aware (Hessian-weighted) compression, from Optimal Brain Damage/Surgeon [16, 17] to Hessian-aware quantization and GPTQ [18, 19]. What is missing across these is a single, *identifiable geometric* object that names the task's metric and lets one predict, recover, and test it. Observation Theory supplies it: the read operator $P_{\mathcal{C}}$, the consumer's local output metric, sitting as the geometric middle term between Shannon's Gaussian Euclidean-MSE specialization ($P_{\mathcal{C}} = I$) and the information bottleneck [12] (an arbitrary, non-geometric relevance variable). We contribute (i) the map placing these literatures on one axis; (ii) $P_{\mathcal{C}}$'s *identifiability*—recovery from black-box consumer queries under a specified observation distribution, validated blind against a planted ground truth, which the quantization literature (estimating the Hessian from calibration data) never validates against truth; (iii) an ex-ante extreme-value reliability certificate; (iv) budget-relativity of the verdict; and (v) a registration-first empirical program—sealed predictions, ex-ante bars, a refuted face carried as prominently as the wins—whose recurring demonstration is a single code whose downstream verdict *inverts* when nothing changes but the consumer. The three companion works are Paper I (the angular observer and the spectral budget-inversion mechanism) [20], Paper II (geometric key quantization and the reconstruction-is-not-the-target negative) [21], and the claim ledger [22]; this is their information-theoretic synthesis.

2 The controlling quantity

Definition 1 (Instance). *An instance is $(X, \mathcal{C}, \mathcal{Q})$: a representation with vectors $x \in \mathbb{R}^d$, a* consumer *$\mathcal{C} : \mathbb{R}^d \to \mathbb{R}^m$ with a divergence D_Y on its outputs, and a family $\mathcal{Q}$ of lossy codes producing $\hat{x} = x + \delta$. The downstream distortion is the pointwise random quantity $d_{\mathcal{O}} := D_Y(\mathcal{C}(x), \mathcal{C}(\hat{x}))$, measured on the consumer output, never assumed from the ambient metric of X; write $\bar{d_{\mathcal{O}}} := \mathbb{E}[d_{\mathcal{O}}]$ for its expectation over the instance.*

Proposition 1 (Consumer-projected distortion). *Let D_Y be twice differentiable with $D_Y(u,u) = 0$, $\nabla_v D_Y(u,v)|_{v=u} = 0$, and* local output metric *$G(u) = \nabla_v^2 D_Y(u,v)|_{v=u} \succeq 0$ (the Fisher information metric when D_Y is the KL divergence or a kindred statistical divergence, a generic local metric otherwise); let $J(x) = \partial\mathcal{C}/\partial x$. For error $\delta = \hat{x} - x$ with conditional* second moment *$\Sigma_\delta(x) = \mathbb{E}[\delta\delta^\top \mid x]$, the leading-order expected distortion is*

$$\bar{d_{\mathcal{O}}} \;=\; \tfrac{1}{2}\,\mathbb{E}_x \operatorname{tr}\big(P_{\mathcal{C}}(x)\,\Sigma_\delta(x)\big) \;+\; r, \qquad P_{\mathcal{C}}(x) = J(x)^\top G(\mathcal{C}(x))\,J(x) \succeq 0,$$

so $P_{\mathcal{C}}$ is the pullback of the local output metric G along $\mathcal{C}$. Making the asymptotics explicit: for a small-noise family $\delta_\varepsilon = \varepsilon Z$ with $\mathbb{E}\|Z\|^2 < \infty$ and the second-order Taylor remainder dominated uniformly on a neighborhood of the diagonal, $\bar{d_{\mathcal{O}}} = \frac{\varepsilon^2}{2}\mathbb{E}[Z^\top P_{\mathcal{C}}(X) Z] + o(\varepsilon^2)$ (i.e. $r = o(\varepsilon^2)$).

When $\Sigma_\delta(x)$ is uncorrelated with the point-dependent weight $P_{\mathcal{C}}(x)$ this factors as $\frac{1}{2}\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ with $P_{\mathcal{C}} = \mathbb{E}_x[P_{\mathcal{C}}(x)]$, $\Sigma_\delta = \mathbb{E}_x[\Sigma_\delta(x)]$. The controlling object is the second *moment, not the covariance: pointwise* $\operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x)) = \operatorname{tr}(P_{\mathcal{C}}(x)\operatorname{Cov}[\delta \mid x]) + \mu_\delta(x)^\top P_{\mathcal{C}}(x)\mu_\delta(x)$ *with* $\mu_\delta(x) = \mathbb{E}[\delta \mid x]$*, then averaged over x, so a biased code's offset is* part *of the distortion, not averaged away—which is why quantizer bias is first-class (§3). Reconstruction distortion $\mathbb{E}\|\delta\|^2 = \operatorname{tr}(\Sigma_\delta)$ is the* isotropic *corner $P_{\mathcal{C}} = I$.*

Proof. Expand $\mathcal{C}(x+\delta) = \mathcal{C}(x) + J\delta + O(\|\delta\|^2)$ and D_Y to second order about its minimum; the value and gradient terms vanish, leaving $\frac{1}{2}\delta^\top(J^\top GJ)\delta + o(\|\delta\|^2)$. Take $\mathbb{E}_x$; the factored form requires the stated uncorrelatedness (for real quantizers both J and Σ_δ depend on x, so the exact object is the x-averaged trace). □

We keep $\mathbb{E}_x \operatorname{tr}(P_{\mathcal{C}}(x)\Sigma_\delta(x))$ as primary and use the factored $\operatorname{tr}(P_{\mathcal{C}}\Sigma_\delta)$ as the convenience it is. (We drop the harmless $\frac{1}{2}$.)

Distinguishability, and the observer's quotient

The form $\delta^\top P_{\mathcal{C}}(x)\delta = (J\delta)^\top G\,(J\delta)$ is not, at root, a compression cost; it is a *distinguishability.* It measures how far the consumer's outputs $\mathcal{C}(x)$ and $\mathcal{C}(x+\delta)$ separate in the output metric G, and when G is the Fisher information metric of the output distribution, $\delta^\top P_{\mathcal{C}}\delta$ *is* the local statistical distinguishability (the leading KL / χ^2) of those outputs. So $P_{\mathcal{C}}$ is the pullback through $\mathcal{C}$ of the geometry of distinguishability: Observation Theory is pullback geometry.

Two consequences follow, and both recur below. First, define the global operational equivalence $x \sim_{\mathcal{C}} x' \iff D_Y(\mathcal{C}(x), \mathcal{C}(x')) = 0$ (statistical indistinguishability of the outputs). Its infinitesimal shadow is $\ker P_{\mathcal{C}}(x) = \ker\big(G^{1/2}J\big)$, which equals $\ker J$ when $G \succ 0$ on $\operatorname{im} J$ but is larger when G is only semidefinite—the output can move parametrically yet stay indistinguishable. Under a constant-rank / integrability condition on $P_{\mathcal{C}}(x)$ this null distribution integrates to leaves tangent to $\ker P_{\mathcal{C}}(x)$; that the *global* divergence vanishes along each leaf—so the leaves are exactly the fibers of $\sim_{\mathcal{C}}$—needs a further compatibility condition, namely $D_Y(\mathcal{C}(x), \mathcal{C}(x')) = 0$ along them (equivalently, $\mathcal{C}$ factors through $X/{\sim_{\mathcal{C}}}$). Where it holds, the consumer's true object is the quotient $X/{\sim_{\mathcal{C}}}$, not X: perturbations in $\ker P_{\mathcal{C}}$ exist in the representation but have *no operational consequence.* In the fixed-$P_{\mathcal{C}}$ model of §8 this quotient is represented by ΠX and Prop. 7 gives the zero-distortion rate $R_{\mathcal{C}}(0) = H(\Pi X)$ (the point-dependent case is open)—one pays only for what one can tell apart. Second, $\ker P_{\mathcal{C}}$ is gauge-like *but only relative to* $\mathcal{C}$: another consumer may read those directions, so it is not a true gauge redundancy (which would be invisible to every admissible observer). One observer's noise is another's signal; one observer's state variable is another's gauge freedom. Adjoining the budget of §7, an observer is a triple $\mathcal{O} = (\mathcal{C}, G, B)$—readout, output metric, and resolution—and the geometry it resolves is relative to all three.

> **Principle of observational relativity.** Operational fidelity, distortion, and resolved geometry are not properties of the object alone; they are properties of the relation among object, observer $(\mathcal{C}, G)$, channel, and budget, and are ill-posed until all are specified. Observation induces geometry; each observer pulls back a world.

The rest of the paper derives $P_{\mathcal{C}}$ for concrete consumers, shows it recoverable from queries, and studies what the channel (§6) and budget (§7) do to the geometry it induces.

3 The read operator, derived

Proposition 2 (Softmax attention)**.** *Read keys $\{k_s\}_{s=1}^S$ through a query bank: logits $l_s = q^\top k_s/\sqrt{d}$, output $p = \mathrm{softmax}(l)$, $D_Y = \mathrm{KL}$. Under key perturbation δ_s,*

$$\bar{d_{\mathcal{O}}} = \tfrac{1}{2d}\,\mathbb{E}_q\,\mathrm{Var}_s^p\big(q^\top\delta_s\big) + O(\|\delta\|^3) = \tfrac{1}{2d}\,\mathbb{E}_q\,\mathrm{tr}\big(qq^\top A_q\big) + O(\|\delta\|^3), \qquad A_q = \sum_s p_s(q)\,\tilde\delta_s\tilde\delta_s^\top,$$

where $\tilde\delta_s = \delta_s - \sum_{s'} p_{s'}\delta_{s'}$ is the across-token p-demeaned error and A_q its $p(q)$-weighted second moment. This factors *to $\frac{1}{2d}\,\mathrm{tr}(Q\,\bar\Sigma_\delta)$, with $Q = \mathbb{E}_q[qq^\top]$ and $\bar\Sigma_\delta = \mathbb{E}_q[A_q]$,* only *when A_q is uncorrelated with $qq^\top$ (the proviso of Prop. 1); in general the expectation stays coupled, since p depends on q.*

Proof. The Fisher metric of softmax is $G = \mathrm{diag}(p) - pp^\top$, so $\mathrm{KL}(p\|\mathrm{softmax}(l+\Delta l)) = \frac{1}{2}\mathrm{Var}_s^p(\Delta l_s) + O(\Delta l^3)$ with $\Delta l_s = q^\top\delta_s/\sqrt{d}$. The $-pp^\top$ term is the softmax's invariance to a common logit shift; it removes the p-mean, giving $\tilde\delta_s$. Since p depends on q, the outer $\bar\Sigma_\delta$ carries the $\mathbb{E}_q$ and the final Q-factorization inherits the uncorrelatedness proviso of Prop. 1. □

Two consequences are the phenomena we later measure. The feature-space factor is generated by $qq^\top$, reducing to $Q = \mathbb{E}_q[qq^\top]$ under the decorrelation / uniform-weight regime above; so an isotropic bank yields an isotropic feature operator (reconstruction-like) while a subspace bank projects—the read operator, hence the winning code, changes with the query distribution. And only the *demeaned* error survives, so the across-token mean is a nuisance the softmax cannot see. This last fact resolves an apparent tension with the DC-offset collapse of Paper II [21]: a truly common error component is free by shift invariance, but a per-channel DC *offset* is not a common component—it inflates the quantizer's range, wasting levels and enlarging the *visible* per-token error, which is exactly why an asymmetric zero-point (asym-NF4) repairs it at no bit cost. Invisible-to-the-consumer and free-to-encode are the same statement; range waste is a different one.

Proposition 3 (Gradient-descent optimizer; exact)**.** *Let $L(\theta) = \frac{1}{2}\theta^\top H\theta$ with $H \succeq 0$, $g = H\theta$, and $\hat g = g + \delta$ with conditionally unbiased error $\mathbb{E}[\delta \mid \theta] = 0$, used in a step of size η. Then $L(\theta-\eta\hat g) - L(\theta-\eta g) = \frac{1}{2}\eta^2\,\delta^\top H\delta - \eta\,\delta^\top H(\theta-\eta g)$, whose error-quadratic part is $\frac{1}{2}\eta^2\,\mathrm{tr}(H\,\delta\delta^\top)$; the linear cross term vanishes in expectation ($\mathbb{E}[\delta \mid \theta] = 0$), giving $\mathbb{E}[d_{\mathcal{O}}] = \frac{1}{2}\eta^2\,\mathrm{tr}(H\,\Sigma_\delta)$, so $P_{\mathcal{C}} = H \succeq 0$* exactly*.*

This is loss-aware pruning/quantization made a metric: Hessian-weighted deletion [16, 17, 18, 19] *is* $\mathrm{tr}(H\Sigma_\delta)$, and calibration-based Hessian estimation is the production form of the probe of §5. Reconstruction is the flat case $H = I$. For a non-convex loss the Hessian may be indefinite; the metric reading then takes $P_{\mathcal{C}}$ to be the PSD Gauss–Newton / Fisher curvature in place of H.

Remark (Retrieval)*.* For score-ranking $s(q, x_i) = q^\top x_i$, the inverted-pair fraction is, to leading order for smooth rank statistics, monotone in $\mathrm{tr}(P_{\mathcal{C}}\bar\Sigma_\delta)/\,\mathrm{tr}(P_{\mathcal{C}}\bar\Sigma_0)$, with $P_{\mathcal{C}} = \mathbb{E}[qq^\top]$, $\bar\Sigma_\delta = \mathrm{Cov}_i(\delta_i)$ the across-item error covariance and $\bar\Sigma_0 = \mathrm{Cov}_i(x_i)$ the signal covariance (both demeaned across items); $P_{\mathcal{C}}$ is a subspace projector when queries concentrate. Rank statistics are non-smooth; we treat this instance empirically (§4).

4 Distortion: the consumer is the metric (GO-2)

Proposition 2 predicts that at matched reconstruction the ranking of two codes is set by $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$, hence by the query distribution.

Dissociation. Two 4-bit codes at an audited matched budget (spread 0.19 bit) with *identical* reconstruction (0.0938 vs 0.0934): the whole-vector code is 2.53× worse on softmax-KL (12/12 seeds); a wrong-quotient control is 21× worse at $\sim 5\times$ the reconstruction. Reconstruction is not the controlling quantity (`[demonstrated]`).

Flip. Holding reconstruction fixed and changing only the query bank, the ranking *inverts*: isotropic ⇒ code a better (2.5×, 12/12); subspace ⇒ code b better (12/12). The directly computed $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$ tracks the sign on every seed in both consumers; reconstruction, a consumer-blind scalar, cannot (rank correlation 0.40 vs 1.00). This is Proposition 2 with Q swapped (`[demonstrated]`).

Replication and out-of-sample. The flip replicates on retrieval ranking (different representation, set-valued consumer): identical reconstruction (0.0964), ranking flips with the read subspace (log-advantage −4.70 vs +4.65, 12/12), $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$ tracking, reconstruction blind (`[replicated]`). It generalizes to optimization with $P_{\mathcal{C}} = H$ (Prop. 3): at matched reconstruction the curvature-metric distortion flips between two Hessians (12/12), tracked by $\mathrm{tr}(H\Sigma_\delta)$, reconstruction uninformative (−0.20)—and under $H = I$ reconstruction tracks *perfectly* (1.00), the derived $P_{\mathcal{C}} = I$ corner observed (`[predicted]`, out-of-sample). One derivation; three read operators. These consumers are *synthetic*—the point is to validate the theory's logic (the $H = I$ positive control being the clean confirmation); the trained-model hardening (§11) is the outstanding validation.

5 Identifiability: the read subspace is recoverable (GO-1)

Proposition 4 (Recovery of the read subspace). *Let $\mathcal{C}(x) = f(Wx)$, $W \in \mathbb{R}^{r\times d}$ of rank r, f coordinatewise with $\mathbb{E}[f_i'(Wx)^2] > 0$ for every coordinate i (no saturated unit). Then $\mathbb{E}_x[J^\top J] = W^\top \mathbb{E}[\mathrm{diag}(f'^2)]W$ has rank r with column space $\mathrm{row}(W)$, so its top-r eigenvectors span the read subspace; a finite-difference estimate $\hat{J}u \approx (\mathcal{C}(x+hu) - \mathcal{C}(x-hu))/2h$ recovers it from consumer queries alone.*

We built such a probe—receiving only the callable $\mathcal{C}(\cdot)$ and the dimensions, never W, the codes, or $d_{\mathcal{O}}$ (blinding enforced in code). It recovers the planted read subspace at overlap 0.936 against a 0.059 chance baseline (≈ 16×), and the *blind* $\mathrm{tr}(\hat{P}_{\mathcal{C}}\Sigma_\delta)$ predicts the flip of §4 on 12/12 seeds in both consumers with rank correlation 1.00—identical to ground truth—while reconstruction cannot (0.40) (`[predicted]`). Two scope notes. Prop. 4 recovers the read *subspace* for the $f(Wx)$ model with a Euclidean output metric; recovering the *full* operator $\mathbb{E}_x[J^\top GJ]$—its eigenvalues, and a general G—is the broader program. And $P_{\mathcal{C}}$ depends on the probe-point distribution and on G, so it is identifiable from black-box consumer queries *under a specified observation distribution*, not from the callable alone. Loss-aware compression estimates such a metric from calibration data; the contribution here is validating *recovery against a planted ground truth under enforced blinding*, and composing it: from consumer queries under that distribution one predicts which code wins, before compressing anything.

6 Capacity: a vacuity threshold, derived (GO-3)

Proposition 5 (Retrieval vacuity threshold). *Model the N–1 distractor scores, standardized to mean 0, variance 1, as i.i.d. standard Gaussian (a reference model). Since* $\Pr[M_{N-1} \le m] = \Phi(m)^{N-1}$, *top-1 success* $\Pr[\hat{\mu} > M_{N-1}]$ *crosses $\frac{1}{2}$ at the* exact median *of the maximum,*

$$\mu_{\mathrm{crit}} := \mathrm{med}(M_{N-1}) = \Phi^{-1}\big(2^{-1/(N-1)}\big), \qquad \textit{leading asymptotic } \sqrt{2\ln(N-1)},$$

where $\hat{\mu} = (\overline{true} - \overline{distr})/\mathrm{std}$ *is the margin certificate and* $\rho = \hat{\mu}/\mu_{\mathrm{crit}}$*; the certificate is* vacuous *for* $\rho < 1$.

Derivation. $\Pr[M_{N-1} \le m] = \Phi(m)^{N-1}$; setting it to $\frac{1}{2}$ gives $m = \Phi^{-1}(2^{-1/(N-1)})$, the exact median. The Gumbel centering is $\alpha_N = \sqrt{2\ln(N-1)} - b_N(\ln\ln(N-1) + \ln 4\pi)/2$ with scale $b_N = 1/\sqrt{2\ln(N-1)}$; the crude $\sqrt{2\ln(N-1)}$ omits the $-b_N(\ln\ln(N-1) + \ln 4\pi)/2$ correction and so *overshoots*. Median and mean share this centering and differ at order b_N: $\mathrm{med}(M_{N-1}) = \alpha_N - b_N \ln\ln 2 + o(b_N)$ and $\mathbb{E}[M_{N-1}] = \alpha_N + \gamma b_N + o(b_N)$ (γ Euler–Mascheroni), so the mean exceeds the median by $(\gamma + \ln\ln 2)\, b_N$. The recall-$\frac{1}{2}$ crossing is the exact median $\Phi^{-1}(2^{-1/(N-1)})$—not the mean, not the leading order. □

Across 14 structurally distinct corpora the certificate orders them by recall at rank correlation 0.991, beating the un-normalized margin (0.873), with a finite-width transition (`[demonstrated]`). The registered harness used the *mean* $\mathbb{E}[M_{N-1}]$ (Monte-Carlo) as its μ_{crit} reference, and the recall-$\frac{1}{2}$ crossing sits at $\rho = 0.948$—slightly *below* 1 against that reference. This is expected and quantitative: the exact $\frac{1}{2}$ threshold is the *median*, below the mean by $\Theta(1/a_N)$, so the $\sim$5% shortfall is largely the mean–median offset (against the exact median of this proposition the crossing sits at ≈ 1). Positive dependence among distractors *can* lower it further by reducing the effective count; heavier tails, by contrast, *raise* the maximum—the two must not be grouped. The exact $\frac{1}{2}$ crossing further assumes a *fixed* standardized true margin and i.i.d. Gaussian distractors; when the true score is itself random or correlated with the distractors, ρ is a reference certificate rather than an exact recall threshold. Capacity says, from a computable quantity, *where* the channel dies before any retrieval is run.

7 Rate: the verdict belongs to the budget (GO-4)

The read operator, capacity, and distortion are stated at a fixed observer budget; the budget is a rate, and it inverts the verdict. The mechanism is spectral: an eigenfunction φ_k with eigenvalue λ_k oscillates on the wavelength $\lambda_k^{-1/2}$, and Weyl's law $N(\lambda) \sim c\,\mathrm{Vol}(M)\lambda^{d_M/2}$ (d_M the intrinsic dimension) grows the count of modes below a fixed scale with resolution.

Lemma 1 (Fixed-band consistency)**.** *At a fixed low-mode budget m, as the sample grows the graph operator converges spectrally to the manifold operator [14], so the fixed low-λ band sharpens and geodesic fidelity* converges toward its continuum value *and is predicted to improve asymptotically (the spectral-convergence mechanism of Paper I [20]). Convergence does not force step-wise monotonicity in N; the monotone finite-sample trend below is the* registered empirical *result, not a consequence of the lemma.*

Holding instead the resolved scale τ fixed (spectral-gap matching), $m = N(\tau)$ grows with N; the newly resolved modes near τ have wavelength approaching the sampling scale and, for a non-ideal substrate, inject sub-geodesic detail, so geodesic fidelity is nonincreasing—Paper I's registered inversion [20]. We replicate it on a new substrate: 768-dimensional multilingual book-paragraph embeddings (Project Gutenberg translations; 32,000 points, three seeds). Fixed $m = 10$: geometry preservation rises with N (trend $+0.4, +1.0, +1.0$). Spectral-gap-matched, $m : 10 \to 121/126/159$: geometry *collapses* (trend -1.0 every seed); the budgets agree at the smallest N (gap ≤ 0.002) (`[replicated]`). Opposite verdicts about the same data, set by the observer's rate: "clean geometry" is a relation between observer resolution and substrate scale, not a substrate invariant.

8 A consumer-relative rate–distortion function

The consumer-projected distortion is the single-letter measure $d_{\mathcal{C}}(x, \hat{x}) = (x-\hat{x})^\top P_{\mathcal{C}}(x-\hat{x})$ for a *fixed* $P_{\mathcal{C}}$ (constant, or the average $\mathbb{E}_x[P_{\mathcal{C}}(x)]$ under the proviso of Prop. 1); define $R_{\mathcal{C}}(D) = \inf\{I(X;\hat{X}) : \mathbb{E}\, d_{\mathcal{C}}(X, \hat{X}) \le D\}$. This fixed-$P_{\mathcal{C}}$ model must be kept distinct from the *local* observation metric $P_{\mathcal{C}}(x)$ of Prop. 1: for a nonlinear consumer $P_{\mathcal{C}}(x)$ varies, there is no single tilted source—a direction invisible at one point can be visible at another—and the coding theorem for that case is open. The results below, and Appendix A, are for the fixed weighted-quadratic model, whose theory is classical [4].

Proposition 6 (Gaussian rate–distortion; Berger reduction)**.** *For $X \sim \mathcal{N}(0, \Sigma_x)$, in the basis diagonalizing $P_{\mathcal{C}}^{1/2}\Sigma_x P_{\mathcal{C}}^{1/2}$ with eigenvalues γ_i,*

$$R_{\mathcal{C}}(D) = \sum_i \tfrac{1}{2}\log_2^+ \frac{\gamma_i}{\theta}, \qquad \sum_i \min(\gamma_i, \theta) = D,$$

a reverse water-filling on the $P_{\mathcal{C}}$-weighted source spectrum. For $P_{\mathcal{C}} = I$ this is Shannon's Gaussian $R(D)$ [2]; the estimation dual is Gauss–Markov least squares [3].

Proof. $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta) = \|P_{\mathcal{C}}^{1/2}(x-\hat{x})\|^2$; with $y = P_{\mathcal{C}}^{1/2}x$ this is MSE on $y \sim \mathcal{N}(0, P_{\mathcal{C}}^{1/2}\Sigma_x P_{\mathcal{C}}^{1/2})$, and Shannon water-filling applies. The transform is invertible off $\ker P_{\mathcal{C}}$, on which the consumer is blind and no rate is spent. □

Bits water-fill on directions the consumer reads *and* the source varies, spending nothing on $\ker P_{\mathcal{C}}$—why a reconstruction-optimal code ($P_{\mathcal{C}} = I$ allocation) is beaten by a consumer-matched one when $P_{\mathcal{C}} \neq I$, and why the winner *flips* with $P_{\mathcal{C}}$ (§4). The $H = I$ control confirms it: only at the isotropic $P_{\mathcal{C}} = I$ is reconstruction the right objective, and there it is exactly right.

Proposition 7 (Consumer-lossless floor; discrete source)**.** *Let X be discrete with finite entropy,* Π *the projector onto* $\mathrm{range}(P_{\mathcal{C}})$*. There exist codes with* $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta) = 0$ *at rate*

$$R_{\mathcal{C}}(0) = H(\Pi X) \le H(X), \qquad H(X) - R_{\mathcal{C}}(0) = H(X \mid \Pi X) \ge 0,$$

with equality iff Π *is injective on* $\mathrm{supp}(X)$*; the saving is strict exactly when distinct symbols differ only within the consumer-invisible subspace* $\ker P_{\mathcal{C}}$*.*

Proof. Achievability: $\mathrm{tr}(P_{\mathcal{C}}\Sigma_\delta)$ depends on δ only through $\Pi\delta$; sending ΠX losslessly and reconstructing $(\mathrm{I}-\Pi)X$ by its conditional mode gives $\Pi\delta = 0$, zero distortion, at rate $H(\Pi X)$. *Converse:* zero distortion forces $\Pi\hat{X} = \Pi X$ a.s., so $I(X;\hat{X}) \ge I(\Pi X; \Pi\hat{X}) = H(\Pi X)$. Hence $R_{\mathcal{C}}(0) = H(\Pi X)$; if Π collapses two support points, $H(\Pi X) < H(X)$. □

Remark (Continuous sources). When the tilted source $Y = P_{\mathcal{C}}^{1/2}X$ has a nondegenerate continuous component, lossless-to-the-consumer coding costs infinite rate ($R_{\mathcal{C}}(D) \to \infty$ as $D \to 0$, consistent with Prop. 6); if X lies entirely in $\ker P_{\mathcal{C}}$ then Y is degenerate and the floor is zero. The surviving statement is that at *every* $D > 0$ the water-filling of Prop. 6, equivalently the reduction of Cor. 1, spends zero rate on $\ker P_{\mathcal{C}}$: the operational rate depends only on the r-dimensional tilted source. This lossless-to-the-consumer phenomenon is exactly coding for computing and graph entropy in the discrete world [8, 9], here given a geometric read operator.

These form a *conceptual axis*, not a formal nesting: Shannon rate–distortion ($P_{\mathcal{C}} = I$, the isotropic observer), consumer-relative rate–distortion ($P_{\mathcal{C}}$ a fixed geometric, *identifiable* operator, Props. 6,7), and the information bottleneck [12]—which optimizes a different object (a stochastic relevance variable) and to which the consumer-relative term connects only through the local-Gaussian relevance limit ($Y = \mathcal{C}(X)$ deterministic, second-order). We claim no containment theorem. The middle term is the operational one—$P_\mathcal{C}$ is PSD, water-fillable, and recoverable from the consumer (Prop. 4)—and its coding theorem (achievability and converse, fixed $P_\mathcal{C}$, general source) is Appendix A.

9 A boundary: the refuted quotient (GO-5)

Not every plausible face survives a sealed prediction. In the spectral setting the α=1 anisotropic normalization of a diffusion operator cancels sampling density and recovers density-free geometry [13]: there the density mode lies in $\ker P_\mathcal{C}$. We conjectured this density quotient likewise restores the invariant in a *non-spectral* domain (locally-scaled ADC retrieval, a hubness correction [15]). It does not, across four prospective tests (direct affinity twice, ADC on real embeddings, non-spectral diffusion-distance). The reason is Proposition 6: for a direct inner-product consumer the density is *read*—it is not in $\ker P_\mathcal{C}$—so quotienting it removes signal and manufactures anti-hubs; in the diffusion domain the direction is right (α=1 optimal, recovered non-spectrally) but the effect is under 2% and *not density-specific* (a uniform-density control gains as much). Per its sealed registration the conjecture is `[refuted]`. The sharpened statement—the α=1 cancellation is a property of the diffusion operator and does not transfer to direct observation—is Proposition 6 read backwards: rate is saved on $\ker P_\mathcal{C}$ only when the nuisance actually lies there. The refutation sharpens the theory rather than embarrassing it.

10 Honest negatives and method

Eleven standing negatives bound the theory and are reported with the prominence of the positives. Reconstruction cosine is not a proxy for key quality (Paper II [21], the motivating negative). The flip is observable only for reconstruction-*matched* codes: absent a tie one code Pareto-dominates and reconstruction is not falsified (a prospective miss locating a precondition, consistent with the water-filling of Prop. 6). The projected-variance trace governs the discriminating comparison but is not a complete rank statistic over all codes. The density quotient (§9). Two are methodological confessions: twice a substantive claim passed at 12/12 while an over-strict secondary gate we had written failed, and twice a degenerate substrate was caught only by a regime-coverage gate.

Every empirical claim was governed by a dated, hashed registry entry committed—sealed—before the first measurement it governs; bars were fixed ex ante; misses enter the ledger as misses; amendments only dated and logged before unblinding; prospective claims (§5,7,9) were blinded where feasible. The house rule is that the paper may not assert what the ledger cannot show.

11 Discussion

In one line: Observation Theory identifies operational geometry with the pullback geometry of distinguishability, then makes that geometry recoverable and experimentally consequential. It asks a change of default. Where a pipeline minimizes reconstruction and hopes the task follows, the theory says: recover the consumer's read operator (§5), water-fill bits on $P_\mathcal{C}$ rather than on I (§8), respect the channel's vacuity threshold (§6), and remember the verdict is relative to the observation budget (§7). Against the literatures it unifies—indirect source coding, coding for computing, rate–distortion–perception, task-oriented communication, loss-aware compression—the contribution is making the task's metric a single identifiable geometric object and testing it adversarially. The decisive open validation is not conceptual but empirical: the analytic consumers of §§4–5 are synthetic, and the one run that converts "the logic is validated" into "the theory works where it matters" is the keys experiment on an actual trained attention layer, with the query second moment recovered by the blind probe and the winning quantizer predicted before compression—Gate B for the book. Open theory: the fixed-$P_{\mathcal{C}}$ achievability and converse of $R_{\mathcal{C}}(D)$ we settle in Appendix A, but the *point-dependent* $P_{\mathcal{C}}(x)$ coding theorem—no single tilted source—remains open; whether the flip is exactly $R_{\mathcal{C}}$'s argmin migrating with $P_{\mathcal{C}}$; and the consumer-relative *channel* coding theorem behind §6. Underneath is one principle: an object has no unique operational geometry—each observer $\mathcal{O} = (\mathcal{C}, G, B)$ pulls back one, and the quotient $X/\sim_{\mathcal{C}}$ is the world it can act on. Four faces, one refutation, eleven negatives: a world to map, now with coordinates in the literature.

A The consumer-relative coding theorem

Throughout, $P_{\mathcal{C}}$ is a *fixed* PSD operator (constant, or $\mathbb{E}_x[P_{\mathcal{C}}(x)]$ under the proviso of Prop. 1); the point-dependent $P_{\mathcal{C}}(x)$ of Prop. 1 is out of scope here (§8).

Lemma 2 (Reduction to the tilted source). *Let $Y = P_{\mathcal{C}}^{1/2} X$, on* $\operatorname{range}(P_{\mathcal{C}})$ *of dimension $r = \operatorname{rank} P_{\mathcal{C}}$. Then $d_{\mathcal{C}}(x, \hat{x}) = \|y - \hat{y}\|^2$ with $\hat{y} = P_{\mathcal{C}}^{1/2}\hat{x}$, and $R_{\mathcal{C}}(D) = R_Y(D)$, the MSE rate–distortion function of Y;* $\ker P_{\mathcal{C}}$ *costs zero rate.*

Proof. $d_{\mathcal{C}} = \|P_{\mathcal{C}}^{1/2}(x - \hat{x})\|^2 = \|y - \hat{y}\|^2$. Objective and constraint depend on $\hat{X}$ only through $\hat{Y}$, so by data processing $I(X; \hat{X}) \geq I(Y; \hat{Y})$; the minimizer places all information in Y (a channel $p(\hat{y} \mid y)$, then $\hat{X} = P_{\mathcal{C}}^{-1/2}\hat{Y}$ on the range, conditional mean on $\ker P_{\mathcal{C}}$), attaining equality. Hence the minimum is $R_Y(D)$. □

Theorem 1 (Achievability and converse). *Let X take values in a standard Borel space, $\{X_t\}$ i.i.d. $\sim p_X$, with a measurable reconstruction alphabet and a reference letter $\hat{x}_0$ satisfying $\mathbb{E}\, d_{\mathcal{C}}(X, \hat{x}_0) < \infty$ and $\mathbb{E}\, d_{\mathcal{C}}(X, \hat{x}_0)^2 < \infty$. For every $D > 0$:* (i) *for any $R > R_{\mathcal{C}}(D)$ there are $(2^{nR}, n)$ codes with $\limsup_n \mathbb{E}\, \frac{1}{n}\sum_t d_{\mathcal{C}}(X_t, \hat{X}_t) \leq D$;* (ii) *every code of expected distortion $\leq D$ has rate $\geq R_{\mathcal{C}}(D)$; and $R_{\mathcal{C}}$ is convex and nonincreasing.*

Proof. This is the single-letter rate–distortion theorem for a memoryless source on a standard Borel space with a measurable difference distortion and finite reference-letter moment [4]; we recall it in *information-density* form, which uses no differential entropies and so holds on general alphabets. *Achievability.* Fix $\varepsilon > 0$ and a test channel $p^*(\hat{x} \mid x)$ with $\mathbb{E}\, d_{\mathcal{C}} \leq D$ and $I(X; \hat{X}) \leq R_{\mathcal{C}}(D) + \varepsilon$; let $p_{\hat{X}}$ be the output marginal and $\imath(x; \hat{x}) = \log \frac{dP_{X\hat{X}}}{d(P_X \times P_{\hat{X}})}$ the information density. Draw $\{\hat{X}^n(w)\}_{w=1}^{2^{nR}}$ i.i.d. $\sim \prod_t p_{\hat{X}}$; encode x^n by any w with $\frac{1}{n}\sum_t \imath(x_t; \hat{X}_t(w)) \leq I(X; \hat{X}) + \varepsilon$ *and* $\frac{1}{n}\sum_t d_{\mathcal{C}}(x_t, \hat{X}_t(w)) \leq D + \varepsilon$, else declare failure and reconstruct with the constant $\hat{x}_0^n$. Let $\mathcal{T}$ be the joint event $\{\frac{1}{n}\sum_t \imath \leq I + \varepsilon,\ \frac{1}{n}\sum_t d_{\mathcal{C}} \leq D + \varepsilon\}$; by the weak law $P^n_{X\hat{X}}(\mathcal{T}) \to 1$, so for all but an $o(1)$-probability set $\mathcal{G}_n$ of source sequences the *conditional* probability $P_{\hat{X}^n|X^n=x^n}(\mathcal{T}) \geq 1 - o(1)$. Changing measure from $P_{\hat{X}^n|x^n}$ to the codebook marginal $P_{\hat{X}^n} = \prod_t p_{\hat{X}}$ on $\mathcal{T}$ costs at most $2^{n(I+\varepsilon)}$ (there the density $\prod_t \frac{dP_{\hat{X}|X}}{dP_{\hat{X}}} = 2^{\sum_t \imath} \leq 2^{n(I+\varepsilon)}$), so for each $x^n \in \mathcal{G}_n$ a single random codeword passes the encoding test with conditional probability $\geq 2^{-n(I+2\varepsilon)}(1 - o(1))$. Hence $\Pr[\text{fail} \mid x^n] \leq \big(1 - 2^{-n(I+2\varepsilon)}(1 - o(1))\big)^{2^{nR}} \to 0$ for $R > I + 2\varepsilon$, uniformly over $x^n \in \mathcal{G}_n$, and $\Pr[\text{fail}] \leq \Pr[\mathcal{G}_n^c] + \sup_{x^n \in \mathcal{G}_n} \Pr[\text{fail} \mid x^n] \to 0$. On success the distortion is $\leq D + \varepsilon$; on failure it is $\frac{1}{n}\sum_t d_{\mathcal{C}}(X_t, \hat{x}_0)$, whose contribution to the expectation is $\leq \sqrt{\mathbb{E}[(\frac{1}{n}\sum_t d_{\mathcal{C}}(X_t,\hat{x}_0))^2]}\sqrt{\Pr[\text{fail}]} \to 0$ by Cauchy–Schwarz, finite by the second-moment hypothesis on $\hat{x}_0$. Thus $\limsup_n \mathbb{E}\,\frac{1}{n}\sum_t d_{\mathcal{C}} \leq D+\varepsilon$; let $\varepsilon \to 0$. *Converse.* With $D_t = \mathbb{E}\, d_{\mathcal{C}}(X_t,\hat{X}_t)$, $nR \geq H(W) \geq I(X^n;\hat{X}^n) \geq \sum_t I(X_t;\hat{X}_t) \geq \sum_t R_{\mathcal{C}}(D_t) \geq nR_{\mathcal{C}}(\frac{1}{n}\sum_t D_t) \geq nR_{\mathcal{C}}(D)$, by memorylessness, the definition of $R_{\mathcal{C}}$, Jensen (convexity), and monotonicity. Convexity and monotonicity follow from time-sharing and distortion relaxation. □

Corollary 1. *$R_{\mathcal{C}}(D)$ equals the MSE rate–distortion of the r-dimensional tilted source $Y = P_{\mathcal{C}}^{1/2}X$; for Gaussian X it is Prop. 6; discretely, $R_{\mathcal{C}}(0) = H(\Pi X)$ (Prop. 7). The operational rate to serve a consumer is governed entirely by the source projected into the read subspace—the precise sense in which Shannon's theorem is the corner $P_{\mathcal{C}} = I$, where the read subspace is everything.*

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Reproducibility & provenance. All results resolve to committed registry entries and artifacts in the public companion repository [22]; each registry entry is committed *before* the run it governs, so the evidence that registration preceded experimentation is the *commit ordering* of the public history. We are explicit about its limits: git and hosting timestamps are not cryptographic third-party proof, and a snapshot taken *after* completion preserves the current history but does not retroactively establish timing. Stronger anchoring—an external immutable deposit (Zenodo/OSF) and any contemporaneous third-party timestamps (signed hashes, prior deposits)—accompanies the paper. Computational experimentation and drafting were carried out with an AI research assistant under the registration-first protocol of §10; every ledger row names its verifier and context-freshness.